UTLA TARUN

Pathapatnam, Srikakulam, Andhra Pradesh
9381713688 tarunutla1234@gmail.com LinkedIn GitHub LeetCode

Professional Summary

AI Engineer with 2+ years of experience designing and deploying production-grade AI/ML and Generative AI applications. Hands-on expertise in Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, LangChain, LangGraph, OpenAI APIs, vector databases, and prompt engineering. Experienced in building scalable AI solutions, intelligent automation workflows, and RESTful AI services using Python and AWS. Passionate about developing reliable, production-ready AI systems with a focus on performance, evaluation, and enterprise integration.

Technical Skills

Languages & Backend: Python, Java, FastAPI, Django, REST APIs, Microservices
GenAI / LLM Systems: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, LangGraph, AI Agents, Agentic Workflows, Prompt Engineering, Context Engineering, OpenAI APIs, Function Calling, Tool Calling, Hybrid Retrieval, Reranking, Hugging Face Transformers, Semantic Search
Machine Learning & CV: Machine Learning, Pandas, Deep Learning, PyTorch, TensorFlow, Scikit-learn, OpenCV, ONNX, ArcFace, InsightFace, NLP
Cloud & Serverless: AWS (Lambda, EC2, S3, API Gateway, CloudWatch), Serverless Architecture, Step Functions, Amazon Bedrock
Databases & Vector Stores: PostgreSQL, pgvector MongoDB, Amazon DynamoDB, FAISS, Milvus (Zilliz), ChromaDB, Vector Embeddings
Tools: Docker, Git, GitHub, VS Code, Postman

Experience

FastCollab — Travel and Expense Management Platform **Sep 2025 – Present**
AI Engineer *Hyderabad, Telangana*

- Built and deployed production-grade Retrieval-Augmented Generation (RAG) systems using Large Language Models (LLMs), vector databases, and semantic search to deliver context-aware enterprise AI solutions.
- Developed an AI-powered assistant using LangGraph, OpenAI APIs, RAG, and intent-based routing to automate user support and provide context-aware guidance across enterprise workflows.
- Designed and implemented a travel policy enforcement engine to evaluate flight and hotel search results against corporate rules, dynamically classifying results as in-policy or out-of-policy.
- Built intelligent document automation workflows using OCR and LLMs to extract, validate, and structure invoice and receipt data for downstream business processes.
- Developed scalable serverless AI microservices using AWS Lambda, API Gateway, and reusable Lambda Layers to support asynchronous AI inference and workflow execution.
- Collaborated with product, engineering, and business teams to design, develop, and deploy production-ready AI solutions following Agile development practices.
- Evaluated retrieval quality and LLM responses using context relevance, answer correctness, and hallucination analysis to improve overall system reliability.

T-Machine Software Solutions Pvt Ltd. **Feb 2024 – Aug 2025**
AI Developer *Chennai, Tamilnadu*

- Developed a face authentication system using ArcFace embeddings, FAISS, and Deep Learning techniques, achieving sub-100 ms semantic similarity search for real-time identity verification.
- Designed a dynamic ONNX model loading pipeline for optimized model inference, enabling scalable deployment of AI models across production environments.
- Enhanced face recognition accuracy through OpenCV-based image preprocessing, including alignment, lighting normalization, and blur detection.
- Developed an NLP-based text classification system using Machine Learning techniques to categorize desktop activities into Work, Entertainment, and Communication classes.
- Built NLP classification pipelines using OpenAI embeddings, SVM, and Random Forest models for semantic feature representation, achieving 85% accuracy on multi-class datasets through feature engineering and model evaluation.
- Developed RESTful backend services using Django REST Framework and PostgreSQL to deliver real-time analytics and productivity insights.
- Collaborated with cross-functional engineering teams following Agile methodologies to develop and deploy production-ready AI solutions.

Projects

Agentic RAG & Document Intelligence Platform

- Built a production-grade Retrieval-Augmented Generation (RAG) platform using Large Language Models (LLMs), LangGraph, OpenAI APIs, and vector databases, supporting hybrid retrieval, reranking, query rewriting, and parent-child chunk retrieval for enterprise document intelligence.
- Developed an asynchronous document ingestion pipeline for PDF and DOCX files, extracting text, tables, images, and code blocks to build structured knowledge bases for semantic search.
- Designed LangGraph-based AI agent workflows for query understanding, retrieval planning, tool orchestration, context expansion, and response generation.
- Implemented semantic vector retrieval using Zilliz (Milvus), OpenAI embeddings, and metadata filtering to improve retrieval relevance and contextual grounding.
- Implemented retrieval evaluation, reranking, and hallucination reduction techniques by measuring context relevance, retrieval precision, and answer quality to improve overall LLM response reliability.

Education

Bapatla Engineering College **Aug. 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering (Honors) | CGPA: 8.94/10 *Bapatla, Andhra Pradesh*